

ABSTRACT

This project focuses on the design and development of a **Smart Rechargeable Motion Sensor Light** system intended to reduce unnecessary electrical power consumption caused by lights remaining switched ON in unoccupied areas. The system is designed to provide automatic lighting control using a Passive Infrared (PIR) motion sensor and a 555 Timer IC while maintaining simplicity, low cost, and energy efficiency.

The PIR sensor detects human motion by sensing changes in infrared radiation emitted by the human body. Whenever motion is detected, the PIR sensor generates a trigger signal that activates the 555 timer circuit configured in monostable mode. The timer then switches ON the LEDs for a predefined duration. If additional motion is detected within the delay period, the timer retriggers and keeps the LEDs ON continuously. When no further motion is detected, the LEDs automatically turn OFF after the preset delay time.

The project also includes a transistor-based LED driver circuit for controlling multiple LEDs efficiently and safely. A manual override switch is incorporated into the system to allow continuous lighting operation whenever required. The complete circuit is powered using a rechargeable battery supply along with a Type-B charging module, making the system portable and suitable for emergency as well as household lighting applications.

The hardware implementation was initially tested on a breadboard and later assembled on a perf-board for improved reliability and compactness. A custom cylindrical enclosure was designed and fabricated using 3D printing technology to provide proper mounting and protection for all components including the PIR sensor, LEDs, battery, switches, and control circuitry.

Experimental observations demonstrated that the system successfully detects human motion, automatically controls lighting, reduces unnecessary energy consumption, and improves user convenience. The developed system is compact, low-cost, rechargeable, portable, and suitable for practical smart lighting applications in homes, staircases, corridors, and indoor environments.

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Chapter 1

INTRODUCTION

1.1 Background

In today's world, energy conservation has become a critical concern due to increasing electricity demand and limited natural resources. A significant amount of electrical energy is wasted due to inefficient usage of lighting systems in homes, offices, and public places. Lights are often left ON even when no one is present, leading to unnecessary power consumption.

To overcome this issue, automatic lighting systems have been developed using motion sensors. One of the most commonly used sensors is the Passive Infrared (PIR) sensor, which detects human movement by sensing changes in infrared radiation emitted by the human body. When motion is detected, the sensor generates a signal that can be used to control electrical devices.

In this project, a timer-based control system is implemented using a 555 timer IC. The timer helps in maintaining the light in ON state for a predefined duration after motion detection. Additionally, a manual override switch is included to provide flexibility in operation, allowing the user to keep the lights continuously ON when required.

1.2 Problem Statement

Conventional lighting systems lack automation and rely entirely on manual operation. This leads to several problems such as:

- Lights remaining ON even when rooms are unoccupied
- Increased electricity consumption and higher energy bills
- Lack of convenience in manually controlling lights frequently
- Inefficient utilization of resources

Although PIR sensors are used in automatic systems, they have certain limitations. They can detect only motion and not the presence of a stationary person. As a result, the light turns OFF even if a person is present but not moving, which can cause inconvenience.

Therefore, there is a need for a system that not only automates lighting but also provides flexibility to overcome the limitations of motion detection.

1.3 Objectives

The main objectives of this project are as follows:

- To design and develop an automatic lighting system using a PIR sensor
- To reduce energy consumption by ensuring lights operate only when required
- To implement a time delay mechanism using a 555 timer IC
- To control multiple LEDs using a transistor-based driver circuit
- To provide a manual override switch for continuous operation
- To create a simple, cost-effective, and reliable system suitable for real-world applications

Chapter 2

LITERATURE REVIEW

2.1 Introduction

The development of automatic lighting systems has gained significant importance in recent years due to the increasing need for energy conservation and smart automation. Various technologies have been explored to design systems that can efficiently control lighting based on human presence and movement. This chapter reviews the existing work related to motion-based lighting systems, timer-based control methods, and modern automation techniques relevant to the proposed project.

2.2 Conventional Lighting Systems

Traditional lighting systems rely on manual switches for operation. These systems require human intervention to turn lights ON and OFF, which often leads to inefficiency. In many cases, lights remain ON even when rooms are unoccupied, resulting in unnecessary energy consumption. Such systems are not reliable for energy saving, especially in homes, offices, and public areas.

Manual systems also lack automation and convenience, making them less suitable for modern smart environments.

2.3 PIR Sensor-Based Systems

Passive Infrared (PIR) sensors are widely used in automatic lighting systems for motion detection. PIR sensors detect changes in infrared radiation emitted by the human body. When a person moves within the detection range, the sensor generates an electrical signal that can be used to control lighting devices.

Several low-cost and energy-efficient lighting systems have been developed using PIR sensors due to their simplicity and ease of implementation.

However, PIR sensors have certain limitations:

- They detect only motion and not the continuous presence of a stationary person
- Very slow movements may not be detected properly
- Environmental conditions can affect sensor performance

Despite these limitations, PIR sensors remain one of the most popular choices for automatic lighting applications because of their affordability and low power consumption.

2.4 Timer-Based Control Systems

Timer circuits are commonly used in automation systems to control the operating duration of electrical devices. One of the most widely used components for this purpose is the 555 Timer IC. The IC can operate in different modes such as monostable, astable, and bistable modes.

In this project, the 555 Timer IC is used in monostable mode to provide a delay after motion detection. Once triggered by the PIR sensor, the timer keeps the LEDs ON for a fixed time interval before automatically switching them OFF.

The use of resistor-capacitor (RC) timing networks makes the design simple, reliable, and cost-effective while providing adjustable delay functionality.

2.5 Transistor-Based LED Driver Systems

When multiple LEDs are used in lighting applications, direct driving from sensor outputs or timer ICs becomes difficult due to current limitations. Therefore, transistor-based driver circuits are commonly used to amplify current and safely control multiple LEDs.

In the proposed system, a transistor switching circuit is used to drive multiple LEDs efficiently. The transistor acts as an electronic switch controlled by the output of the 555 timer IC.

This approach offers the following advantages:

- Ability to control multiple LEDs
- Improved current handling capability
- Better reliability and switching performance
- Protection of the timer IC from excessive load

2.6 Rechargeable Lighting Systems

Rechargeable lighting systems are becoming increasingly popular due to their portability and energy efficiency. These systems use rechargeable batteries along with charging modules to provide uninterrupted operation without direct dependency on mains power.

In this project, a rechargeable power supply is used to make the lighting system portable and suitable for emergency and low-power applications.

Rechargeable systems provide advantages such as:

- Portability
- Reduced power consumption
- Reusability of battery power
- Better suitability for compact devices

2.7 Limitations of Existing Systems

Based on the study of existing systems, the following limitations are identified:

- PIR sensors cannot detect stationary human presence continuously
- Many systems do not include manual override functionality
- Advanced automation systems increase cost and complexity
- Some systems require programming knowledge and microcontrollers

These limitations reduce usability and increase implementation complexity in low-cost applications.

2.8 Proposed Approach

To overcome the limitations of existing systems, the proposed project combines a PIR motion sensor with a 555 Timer IC and a transistor-based LED driver circuit. A manual override switch is also included to allow continuous lighting operation whenever required.

The system is designed to be rechargeable, compact, low-cost, and energy-efficient.

The proposed approach provides the following advantages:

- Simple and low-cost implementation
- Automatic lighting control

- Reduced energy wastage
- Adjustable delay using 555 timer IC
- Manual override for continuous operation
- Rechargeable and portable design

Chapter 3

METHODOLOGY

3.1 Block Diagram

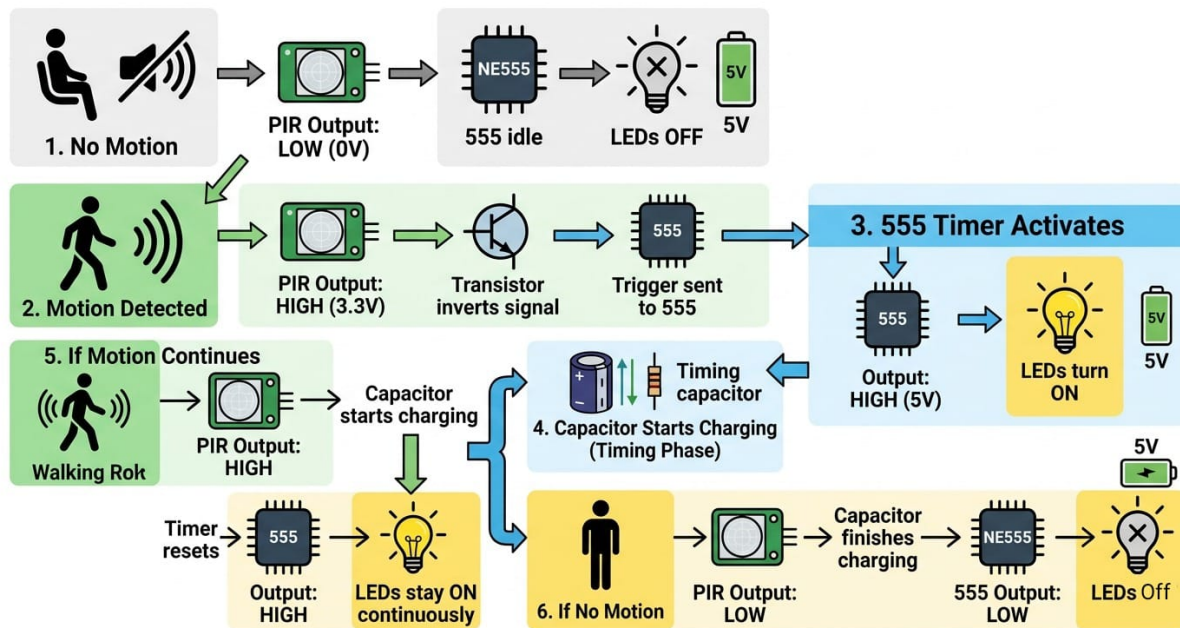


Figure 3.1: Block Diagram of Automatic Light Control System Using PIR Sensor

Explanation of Block Diagram:

1. When there is no motion, the PIR sensor output remains LOW (0V), so the 555 timer stays idle and the LEDs remain OFF.
2. When motion is detected, the PIR sensor produces a HIGH output signal (around 3.3V).
3. This signal is passed through transistor Q1, which acts as a switching stage and pulls the trigger pin of the 555 timer LOW.
4. The 555 timer activates in monostable mode and its output becomes HIGH (5V).
5. As a result, the LEDs turn ON.
6. At the same time, the timing capacitor starts charging through the resistor network, which determines the ON-time of the LEDs.

7. If motion continues, the PIR sensor keeps sending signals, retriggering the timer so the LEDs stay ON continuously.
8. If no motion is detected, the PIR sensor output goes LOW and no new trigger is given.
9. After the capacitor fully charges, the 555 timer output becomes LOW and the LEDs turn OFF.

3.2 Circuit Diagram

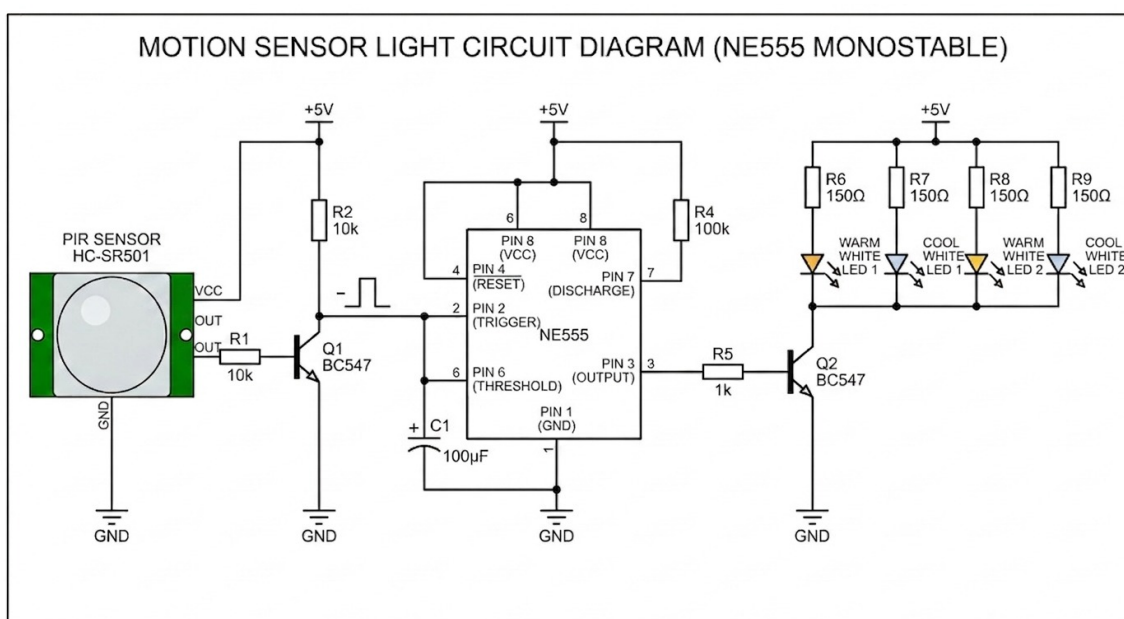


Figure 3.2: Circuit Diagram of Motion Sensor Light (NE555 Monostable)

Explanation of Circuit Diagram:

1. The circuit is powered using a +5V supply which is given to the PIR sensor, 555 timer IC, and LEDs.
2. When there is no motion, the PIR sensor output remains LOW and the LEDs remain OFF.
3. When motion is detected, the PIR sensor gives a HIGH output signal.
4. This signal is applied to transistor Q1, which pulls the trigger pin of the 555 timer LOW and activates the timer.
5. The 555 timer gets activated in monostable mode and its output becomes HIGH.

6. At the same time, capacitor C1 starts charging, setting the ON-time of the LEDs.
7. The output from the 555 timer is given to transistor Q2, which acts as a switch.
8. Q2 turns ON and allows current to flow through the LEDs, causing them to glow.
9. If motion continues, the timer is retriggered and the LEDs stay ON.
10. When no further motion is detected, the timing capacitor continues charging until the preset delay time is completed.
11. After the delay period, the 555 timer output becomes LOW, Q2 turns OFF, and the LEDs turn OFF.

3.3 Working Principle

1. The system is powered using a +5V supply.
2. When there is no motion, the PIR sensor output remains LOW, so the LEDs stay OFF.
3. When motion is detected, the PIR sensor produces a HIGH output signal.
4. This signal is passed through transistor Q1, which acts as a switching stage to trigger the 555 timer.
5. The processed signal triggers the 555 timer IC operating in monostable mode.
6. The 555 timer output becomes HIGH for a fixed time duration.
7. This HIGH output is given to transistor Q2, which acts as a switch.
8. Q2 turns ON and allows current to flow through the LEDs.
9. At the same time, the timing capacitor starts charging through the resistor network, determining the ON-time of the LEDs.
10. If motion continues, the timer is retriggered and the LEDs remain ON continuously.
11. When motion stops, no new trigger is given to the timer.

12. After the preset delay time, the 555 timer output becomes LOW, Q2 turns OFF, and the LEDs turn OFF.

Capacitor Value	LED ON Time
100 μ F	8 seconds
220 μ F	18 seconds
470 μ F	40 seconds
1000 μ F	85 seconds

Table 3.1: Effect of Capacitor Value on LED ON Time

Chapter 4

SYSTEM DESIGN / IMPLEMENTATION

4.1 Introduction

The Smart Rechargeable Motion Sensor Light system is designed to provide automatic lighting control using motion detection technology. The system combines a PIR motion sensor, NE555 Timer IC, transistor switching circuit, rechargeable power supply, and LED lighting arrangement to create an energy-efficient and portable lighting solution.

The implementation of the project involves sensor interfacing, timer-based delay control, transistor-based LED switching, rechargeable battery integration, and enclosure design. The complete system was first tested on a breadboard and later assembled on a perfboard for better reliability and compactness.

4.2 Flowchart

SMART RECHARGABLE MOTION SENSOR LIGHT SYSTEM

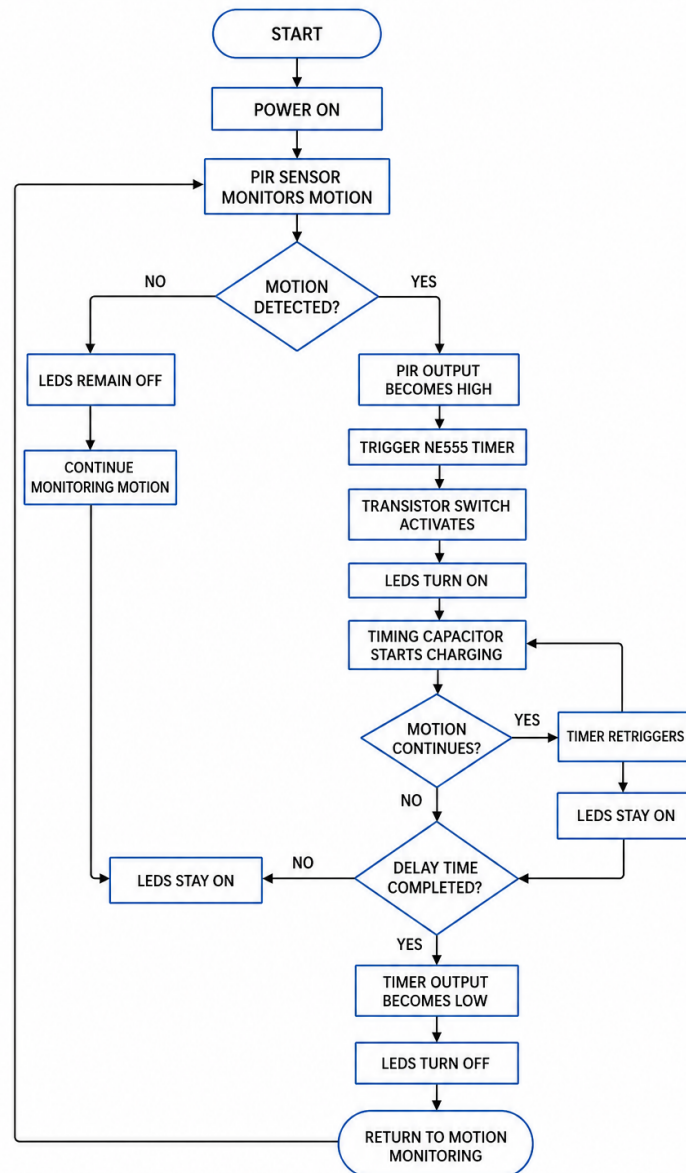


Figure 4.1: Flowchart of Smart Rechargeable Motion Sensor Light System

4.3 System Design

The overall system consists of the following major blocks:

- PIR Motion Detection Unit
- 555 Timer Delay Circuit
- Transistor Switching Circuit
- LED Lighting Unit
- Rechargeable Power Supply
- Manual Override Switch

The PIR sensor continuously monitors the surroundings for human motion. When motion is detected, the sensor generates an output signal that triggers the 555 Timer IC configured in monostable mode. The timer output then activates the transistor switching circuit, which powers the LEDs for a predefined duration.

A manual override switch is also included to allow continuous illumination whenever required.

4.4 Implementation of PIR Sensor Circuit

The HC-SR501 PIR sensor is used for detecting human motion. The sensor operates on a 5V DC supply and provides a HIGH output signal whenever motion is detected within its sensing range.

The PIR sensor output is connected to transistor Q1, which acts as a signal conditioning stage. Q1 helps in properly triggering the 555 timer by pulling the trigger pin LOW whenever motion is detected.

The sensitivity and delay settings of the PIR sensor were adjusted during testing to achieve maximum detection range and reliable operation.

4.5 Implementation of 555 Timer Circuit

The NE555 Timer IC is configured in monostable mode to generate a time delay for LED operation. The timer receives a trigger pulse from the PIR sensor through transistor Q1.

When triggered, the output of the 555 timer becomes HIGH and remains HIGH until the timing capacitor charges to the threshold voltage. The ON-time delay is determined using a resistor-capacitor (RC) timing network.

The timing equation used is:

$$T = 1.1RC$$

where:

- T = Delay time
- R = Timing resistor
- C = Timing capacitor

Different capacitor values were tested experimentally to obtain suitable delay durations.

4.6 Implementation of LED Driver Circuit

The output current capability of the 555 timer is limited when driving multiple LEDs. Therefore, transistor Q2 (BC547) is used as an electronic switching device to control the LED load efficiently.

When the 555 timer output becomes HIGH, transistor Q2 turns ON and allows current to flow through the LEDs. Multiple white LEDs are connected in parallel, each with an individual current-limiting resistor to ensure safe and uniform operation.

The LEDs are arranged in a circular pattern to provide better illumination distribution.

4.7 Implementation of Manual Override Switch

A manual override switch is integrated into the circuit to allow continuous lighting operation whenever required. The switch bypasses the automatic PIR control and directly forces the transistor switching stage into the ON state.

This feature improves usability and allows the system to function both as an automatic and manual lighting device.

4.8 Rechargeable Power Supply Design

The system is powered using a rechargeable battery supply along with a Type-B charging module. The rechargeable power source improves portability and allows the device to operate independently without direct dependence on external power.

The power supply section provides regulated 5V DC output for the PIR sensor, 555 timer IC, transistor circuit, and LEDs.

4.9 Breadboard Testing and Perfboard Assembly

Initially, the complete circuit was implemented and tested on a breadboard to verify circuit operation, timing performance, motion detection, and LED switching behavior.

After successful testing and troubleshooting, the circuit was soldered onto a perfboard for permanent implementation. Proper wire routing and soldering techniques were used to improve circuit reliability and compactness.

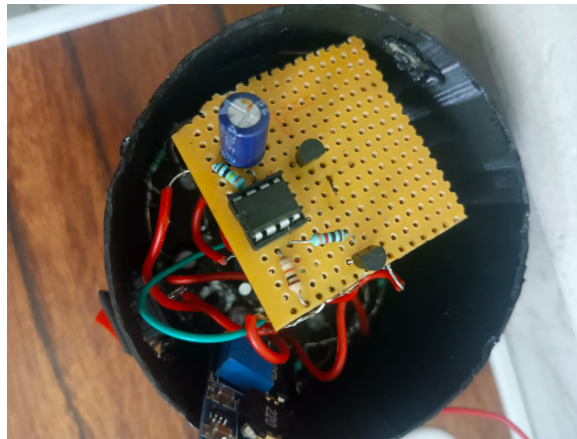


Figure 4.2: Perfboard Assembly of Smart Rechargeable Motion Sensor Light

4.10 3D Printed Enclosure Design

A cylindrical enclosure was designed using 3D modeling software and fabricated using a 3D printer. The enclosure includes:

- Circular openings for LEDs
- Central opening for PIR sensor placement
- Side opening for manual switch
- Space for rechargeable battery and circuit board
- Type-B charging port access

The enclosure improved the appearance, portability, and mechanical protection of the system.

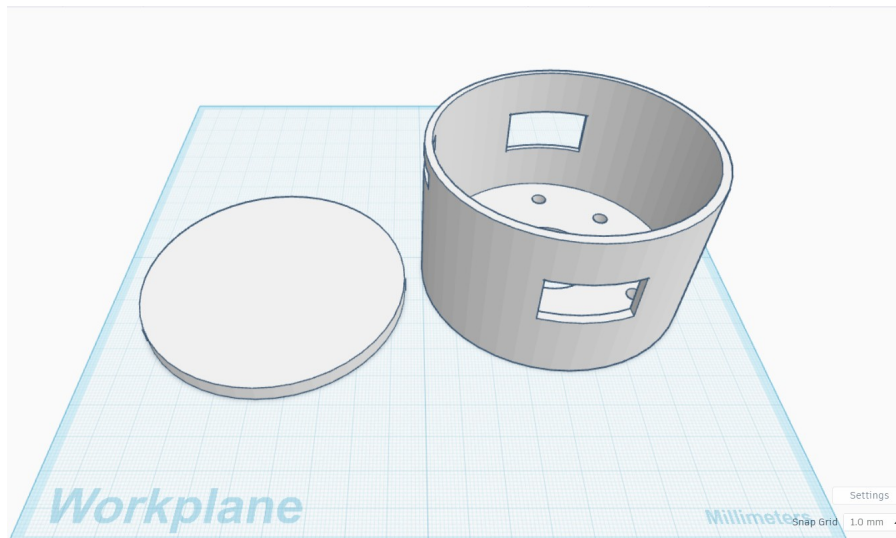


Figure 4.3: 3D Designed Enclosure for Smart Rechargeable Motion Sensor Light

4.11 System Operation

The complete working sequence of the system is as follows:

1. The PIR sensor continuously monitors for motion.
2. When motion is detected, the PIR sensor generates a HIGH output signal.
3. Transistor Q1 triggers the 555 timer IC.
4. The 555 timer output becomes HIGH for a predefined duration.
5. Transistor Q2 turns ON and powers the LEDs.
6. The LEDs remain ON until the timing capacitor completes charging.
7. If additional motion is detected, the timer retriggers and extends the ON duration.
8. If no motion is detected, the timer output becomes LOW after the delay period and the LEDs turn OFF.
9. The manual switch can force the LEDs to remain continuously ON whenever required.

Chapter 5

RESULTS AND DISCUSSION

The implemented Smart Rechargeable Motion Sensor Light system successfully detects motion and automatically controls LED illumination. The following outputs and observations were obtained during the testing and implementation of the project:

- LEDs remain OFF when no motion is detected.
- LEDs turn ON immediately after detecting motion.
- LEDs stay ON for the preset delay period.
- LEDs automatically turn OFF after inactivity.
- The manual override switch keeps the LEDs continuously ON whenever required.

5.1 Output Screenshots

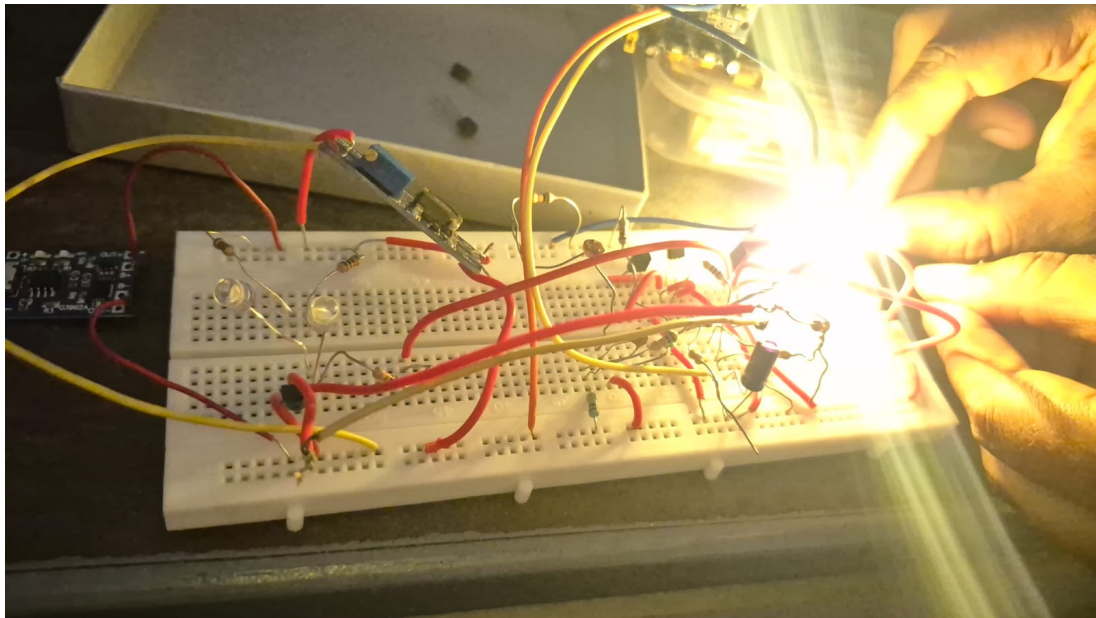


Figure 5.1: Breadboard Testing of Smart Rechargeable Motion Sensor Light



Figure 5.2: Final Front View of the Smart Rechargeable Motion Sensor Light

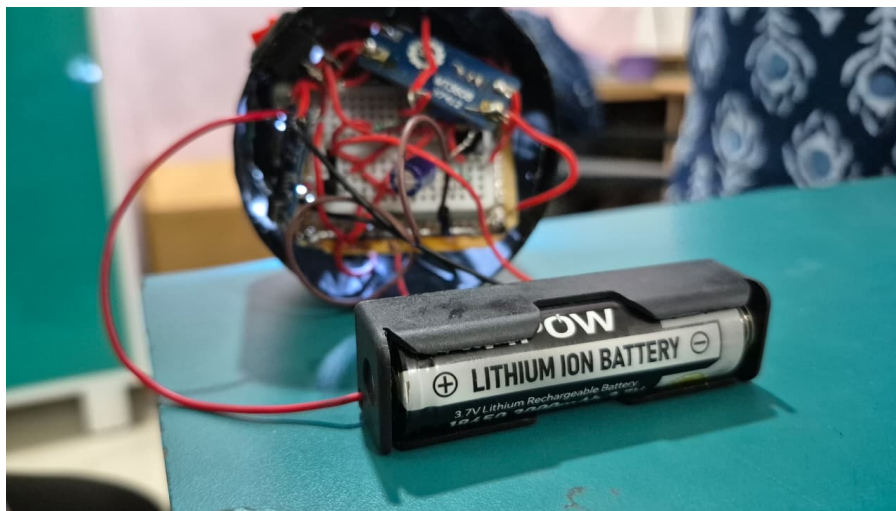


Figure 5.3: Internal Hardware Assembly with Rechargeable Battery



Figure 5.4: 3D Printing Process of the Enclosure

The final assembled prototype consists of:

- 8 white LEDs mounted in a circular pattern
- Central PIR motion sensor for motion detection
- Manual rocker switch for control operation
- Internal perfboard circuit containing the NE555 timer IC, BC547 transistors, resistors, and capacitors
- Rechargeable lithium-ion battery power supply

Distance from Sensor	Detection Status
1 meter	Strong Detection
2 meters	Strong Detection
3 meters	Strong Detection
4 meters	Strong Detection
5 meters	Moderate Detection
6 meters	Weak Detection
7 meters	Maximum Detectable Range
Beyond 7 meters	Not Detected

Table 5.1: PIR Motion Detection Range Observation

The 3D printed enclosure improved:

- Mechanical protection of components
- Compactness and portability
- Ease of handling and installation
- Overall aesthetic appearance of the project

The fabricated prototype operated successfully and demonstrated efficient automatic lighting control based on motion detection along with manual override functionality.

5.2 Analysis

The Smart Rechargeable Motion Sensor Light system was successfully designed, implemented, and tested for automatic lighting applications. The system effectively detects human motion using the PIR sensor and automatically controls LED illumination based on the detected movement.

The HC-SR501 PIR sensor demonstrated reliable motion sensing capability within a short response time. Whenever motion was detected, the sensor generated a trigger signal for the NE555 Timer IC, which maintained the LEDs in the ON state for a predefined duration. The timer operation was stable and consistent throughout the testing process.

The use of BC547 transistors improved signal processing and enabled efficient switching of the LED load. The LEDs produced sufficient brightness for night illumination while maintaining low power consumption.

The implemented system provides several advantages:

- Automatic operation without manual intervention
- Energy-efficient lighting control
- Low power consumption during idle conditions
- Simple and low-cost circuit design
- Compact and portable structure
- Easy installation and maintenance
- Reliable switching operation
- Adjustable delay timing using resistor-capacitor combinations

The inclusion of a manual override switch increased the usability of the system by allowing continuous illumination whenever required. This feature makes the device suitable for both automatic and manual lighting applications.

The custom 3D printed enclosure improved the overall appearance and functionality of the project. The casing securely housed the PIR sensor, LEDs, switch, battery supply, and circuit components in a compact cylindrical structure. The circular arrangement of LEDs provided uniform light distribution, while the central positioning of the PIR sensor improved motion detection coverage.

The rechargeable lithium-ion battery supply enhanced portability and allowed the device to operate independently without requiring a direct external power connection. The Type-B charging module also improved charging convenience and modernized the power system design.

During testing, the following observations were made:

- The PIR sensor accurately detected moving human objects.
- LED response was immediate after motion detection.
- Delay timing depended on the capacitor value used in the 555 timer circuit.
- Larger capacitor values increased the LED ON duration.
- The system remained stable during continuous operation.

- The manual switch operated correctly for override functionality.

However, certain limitations were observed in the system:

- The PIR sensor detects only motion and cannot detect stationary human presence.
- False triggering may occur due to sudden temperature changes or nearby heat sources.
- Detection range and sensitivity are affected by sensor positioning.
- Battery backup duration depends on LED brightness and operating time.

Overall, the Smart Rechargeable Motion Sensor Light system demonstrated efficient, reliable, and cost-effective performance for automatic lighting control applications. The project successfully combined motion sensing, timer control, manual override operation, rechargeable power supply, and 3D printed enclosure design into a compact and functional smart lighting solution.

5.3 Cost Estimation

Sr. No.	Component	Quantity	Approx. Cost (INR)
1	HC-SR501 PIR Sensor	1	50
2	NE555 Timer IC	1	8
3	BC547 Transistors	2	20
4	White LEDs	8	20
5	Resistors and Capacitors	Multiple	10
6	Perfboard / PCB	1	10
7	Type-B Power Module	1	16
8	Rechargeable Battery	1	35
9	Switches and Wires	Multiple	10
10	3D Printed Enclosure	1	(from college)
Total Estimated Cost			179 INR

Table 5.2: Approximate Cost Estimation of the Project

The project was developed using low-cost and easily available electronic components, making it suitable for practical household and small-scale automation applications.

Chapter 6

CONCLUSION

The Smart Rechargeable Motion Sensor Light system was successfully designed and implemented using a PIR motion sensor, NE555 Timer IC, BC547 transistors, LEDs, rechargeable battery supply, and a 3D printed enclosure. The system automatically controls lighting based on human motion, thereby reducing unnecessary power consumption and improving energy efficiency.

The PIR sensor effectively detected human movement and triggered the lighting system, while the NE555 Timer IC provided stable delay operation for automatic LED control. The transistor switching circuit ensured reliable operation of the LEDs, and the manual override switch added flexibility by allowing continuous lighting whenever required.

The project demonstrated several important features such as automatic lighting control, low power consumption, compact design, portability, and cost-effective implementation. The use of a rechargeable lithium-ion battery and Type-B charging module further improved the practicality and usability of the system.

The custom 3D printed casing provided proper housing for all electronic components and improved the overall appearance, safety, and durability of the device. The circular arrangement of LEDs and central placement of the PIR sensor ensured effective illumination and motion detection coverage.

This project also helped in gaining practical knowledge of electronic circuit design, sensor interfacing, timer circuits, hardware assembly, soldering, troubleshooting, PCB planning, and 3D printing technology.

Overall, the Smart Rechargeable Motion Sensor Light system proved to be an efficient, reliable, and user-friendly solution for automatic lighting applications in homes, staircases, corridors, and indoor environments.

Advantages	Limitations
Low power consumption	PIR cannot detect stationary humans
Automatic operation	Sensitive to heat sources
Low-cost implementation	Limited detection angle
Portable and rechargeable	Battery backup depends on usage

Table 6.1: Advantages and Limitations of the System

Chapter 7

FUTURE SCOPE

The Smart Rechargeable Motion Sensor Light system can be further improved and enhanced in the following ways:

1. Advanced Motion Detection

The PIR sensor can be replaced with a microwave radar sensor or presence sensor for improved human detection, including detection of stationary persons.

2. Brightness Control System

PWM (Pulse Width Modulation) techniques can be implemented to control LED brightness according to environmental conditions and user requirements.

3. Improved Battery Backup

A higher-capacity rechargeable battery system can be integrated to provide longer backup duration and uninterrupted operation.

4. Solar Power Integration

Solar charging functionality can be added to make the system more energy-efficient and environmentally friendly.

5. Microcontroller-Based Automation

A microcontroller such as Arduino Nano or ESP32 can be used for advanced automation, programmable timing control, and intelligent operation.

6. Ambient Light Sensing

An LDR (Light Dependent Resistor) can be added to enable automatic day/night operation, ensuring that the light works only in low-light conditions.

7. IoT-Based Smart Control

Wireless technologies such as Wi-Fi or Bluetooth can be integrated for remote monitoring and smartphone-based control.

8. Compact PCB Design

A professionally designed PCB can replace the perfboard circuit to improve reliability, compactness, and overall appearance.

9. Enhanced 3D Printed Enclosure

The enclosure design can be further improved for better durability, waterproofing, heat management, and aesthetics.

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